

1. 判别下列广义积分的敛散性;若收敛,求出其值:

(1) $\int_0^{+\infty} x e^{-x} dx$; (2) $\int_0^{+\infty} \frac{dx}{(x+1)(x+2)}$;

(3) $\frac{1}{\sigma \sqrt{2\pi}} \int_{-\infty}^{+\infty} e^{-\frac{(x-u)^2}{2\sigma^2}} dx$ ($\sigma > 0$);

1. (1) $\int_0^{+\infty} x e^{-x} dx = \int_0^{+\infty} -x d e^{-x} = -x e^{-x} \Big|_0^{+\infty} + \int_0^{+\infty} e^{-x} dx$
 $= 0 - e^{-x} \Big|_0^{+\infty} = 1$

$\therefore$ 收敛且值为1

(2) $\int_0^{+\infty} \frac{dx}{(x+1)(x+2)} = \int_0^{+\infty} \frac{dx}{x+1} - \int_0^{+\infty} \frac{dx}{x+2} = \ln \frac{x+1}{x+2} \Big|_0^{+\infty} = \ln 2$.

$\therefore$ 收敛.

(4) $\int_0^{+\infty} \frac{1+x^2}{1+x^4} dx$ (提示:做变换 $u = x - \frac{1}{x}$);

(5) $\int_0^{+\infty} x \sin x dx$; (6) $\int_1^{+\infty} \frac{dx}{x \sqrt{x-1}}$;

(7) $\int_0^{+\infty} \frac{\arctan x}{(1+x^2)^{3/2}} dx$; (8) $\int_{-\infty}^{+\infty} \frac{dx}{x^2+2x+2}$;

(9) $\int_0^{+\infty} e^{-x} \cos x dx$; (10) $\int_{-1}^1 \frac{dx}{\sqrt{1-x^2}}$;

(11) $\int_0^{1/2} \frac{dx}{x \ln x}$; (12) $\int_0^{1/2} \frac{dx}{x \ln^2 x}$;

(13) $\int_0^{+\infty} \frac{dx}{1+x^4}$ (提示:令 $x = \frac{1}{t}$, 则原式 $= \int_0^{+\infty} \frac{t^2}{1+t^4} dt$, 由此可推出

原式 $= \frac{1}{2} \int_0^{+\infty} \frac{1+t^2}{1+t^4} dt$.

(5) $\int_0^{+\infty} x \sin x dx = (\sin x - x \cos x) \Big|_0^{+\infty}$

而 $\lim_{x \rightarrow \infty} (\sin x - x \cos x)$ 不存在

故发散

(6) 取 $x = t^2+1$

$\int_4^{+\infty} \frac{dx}{x \sqrt{x-1}} = \int_3^{+\infty} \frac{2t dt}{t \cdot (t^2+1)} = \int_3^{+\infty} \frac{2dt}{t^2+1} = 2 \arctan t \Big|_3^{+\infty} = 2 \cdot (\frac{\pi}{2} - \frac{\pi}{3}) = \frac{\pi}{3}$

(7) $\int_0^{+\infty} \frac{\arctan x}{(1+x^2)^{3/2}} dx = \int_0^{+\infty} \frac{\arctan x}{\sqrt{1+x^2}} d \arctan x$

令 $t = \arctan x \in [0, \frac{\pi}{2}]$

$\therefore x = \tan t$,

$\pi \frac{1}{2} = \int_0^{\frac{\pi}{2}} t \cos t \cdot dt = \int_0^{\frac{\pi}{2}} t ds \sin t = t \sin t \Big|_0^{\frac{\pi}{2}} - \int_0^{\frac{\pi}{2}} \sin t dt = \frac{\pi}{2} - 1$

(8) $\int_{-\infty}^{+\infty} \frac{dx}{x^2+2x+2} = \int_{-\infty}^{+\infty} \frac{d(x+1)}{(x+1)^2+1} = \int_{-\infty}^{+\infty} \frac{dt}{t^2+1} = \arctan t \Big|_{-\infty}^{+\infty} = \frac{\pi}{2} + \frac{\pi}{2} = \pi$

(9) $\int_0^{+\infty} e^{-x} \cos x dx = - \int_0^{+\infty} \cos x d e^{-x} = - (\cos x e^{-x} \Big|_0^{+\infty} + \int_0^{+\infty} e^{-x} \sin x dx) = 1 - \int_0^{+\infty} e^{-x} \sin x dx$

而 $\int_0^{+\infty} e^{-x} \sin x dx = \int_0^{+\infty} e^{-x} d \sin x = e^{-x} \sin x \Big|_0^{+\infty} + \int_0^{+\infty} e^{-x} \sin x dx = \int_0^{+\infty} e^{-x} \sin x dx$

$\therefore \int_0^{+\infty} e^{-x} \cos x dx = \frac{1}{2}$

$$\begin{aligned}
 (9) \int_0^{+\infty} e^{-x} \cos x \, dx; \quad (10) \int_{-1}^1 \frac{dx}{\sqrt{1-x^2}}; \\
 (11) \int_0^{1/2} \frac{dx}{x \ln x}; \quad (12) \int_0^{1/2} \frac{dx}{x \ln^2 x}; \\
 (13) \int_0^{+\infty} \frac{dx}{1+x^4} \quad \left(\text{提示: 令 } x = \frac{1}{t}, \text{ 则原式} = \int_0^{+\infty} \frac{t^2}{1+t^4} dt, \text{ 由此可推出} \right. \\
 \left. \text{原式} = \frac{1}{2} \int_0^{+\infty} \frac{1+t^2}{1+t^4} dt. \right.
 \end{aligned}$$

再用第(4)小题的结果).

2. 证明下列各式($\sigma > 0$):

$$(1) \frac{1}{\sigma \sqrt{2\pi}} \int_{-\infty}^{+\infty} (x-a)^2 e^{-\frac{(x-a)^2}{2\sigma^2}} dx = \sigma^2;$$

$$(2) \frac{1}{\sigma \sqrt{2\pi}} \int_{-\infty}^{+\infty} x e^{-\frac{(x-a)^2}{2\sigma^2}} dx = a.$$

$$(11) \int_0^{\frac{1}{2}} \frac{dx}{x \ln x} = \ln |\ln x| \Big|_0^{\frac{1}{2}}$$

$\lim_{x \rightarrow 0} \ln |\ln x|$ 不存在, $\therefore$ 发散

$$(12) \int_0^{\frac{1}{2}} \frac{dx}{x \ln^2 x} = \int_0^{\frac{1}{2}} \frac{d \ln x}{\ln^2 x} = -\frac{1}{\ln x} \Big|_0^{\frac{1}{2}} = \frac{1}{\ln 2} + \lim_{A \rightarrow 0} \frac{1}{\ln A} = \frac{1}{\ln 2}$$

$$\begin{aligned}
 2. (1) \frac{1}{6\sqrt{2\pi}} \int_{-\infty}^{+\infty} y^2 e^{-\frac{y^2}{2\sigma^2}} dy &= \frac{26^2}{\sqrt{2\pi}} \int_{-\infty}^{+\infty} \left(\frac{y}{\sqrt{2\sigma}}\right)^2 e^{-\left(\frac{y}{\sqrt{2\sigma}}\right)^2} d\frac{y}{\sqrt{2\sigma}} = \frac{26^2}{\sqrt{2\pi}} \int_{-\infty}^{+\infty} t^2 e^{-t^2} dt \\
 &= \frac{46^2}{\sqrt{2\pi}} \int_0^{+\infty} t^2 e^{-t^2} dt
 \end{aligned}$$

$$\begin{aligned}
 \left[\int_0^{+\infty} t^2 e^{-t^2} dt \right]^2 &= \int_0^{+\infty} x^2 e^{-x^2} dx \cdot \int_0^{+\infty} y^2 e^{-y^2} dy \\
 &= \iint_D x^2 y^2 e^{-x^2-y^2} d\sigma, \quad \text{其中 } D \text{ 为第一象限}
 \end{aligned}$$

使用极坐标换元

$$\begin{aligned}
 \text{原式} &= \int_0^{\frac{\pi}{2}} d\theta \cdot \int_0^{+\infty} r^5 \sin^2 \theta \cos^2 \theta \cdot e^{-r^2} dr \\
 &= \left(\frac{\pi}{4} - \frac{3}{16}\pi \right) \cdot \int_0^{+\infty} r^5 \cdot e^{-r^2} dr \\
 &= \frac{1}{16}\pi \cdot \frac{1}{2} \int_0^{+\infty} r^4 e^{-r^2} dr \\
 &= \frac{1}{32}\pi \cdot \int_0^{+\infty} t^2 e^{-t} dt
 \end{aligned}$$

$$\begin{aligned}
 \int_0^{+\infty} t^2 e^{-t} dt &= - \int_0^{+\infty} t^2 de^{-t} = - \left(t^2 e^{-t} \Big|_0^{+\infty} - \int_0^{+\infty} 2t \cdot e^{-t} dt \right) \\
 &= - \int_0^{+\infty} 2t \cdot de^{-t} = -2 \cdot \left(t e^{-t} \Big|_0^{+\infty} + \int_0^{+\infty} e^{-t} dt \right) \\
 &= 2.
 \end{aligned}$$

$$\therefore \text{原式} = \frac{1}{16}\pi, \quad \int_0^{+\infty} t^2 e^{-t^2} dt = \frac{\sqrt{2}}{4}$$

$$\therefore \text{原式} = 6^2$$

$$\begin{aligned}
 (2) \quad \int_{-\infty}^{+\infty} x \cdot e^{-\frac{(x-a)^2}{2b^2}} dx &= \int_{-\infty}^{+\infty} (x+a) e^{-\frac{x^2}{2b^2}} dx = \sqrt{2}b \cdot \int_{-\infty}^{+\infty} (x+a) \cdot e^{-\left(\frac{x}{\sqrt{2}b}\right)^2} d\frac{x}{\sqrt{2}b} \\
 &= \sqrt{2}b \cdot \int_{-\infty}^{+\infty} (\sqrt{2}b \cdot x + a) \cdot e^{-x^2} dx \\
 &= \sqrt{2}b \cdot \left(\int_{-\infty}^{+\infty} \sqrt{2}b \cdot x e^{-x^2} dx + a \int_{-\infty}^{+\infty} e^{-x^2} dx \right) \\
 &= 2b^2 \cdot \int_{-\infty}^{+\infty} x e^{-x^2} dx + \sqrt{2}b \cdot a \sqrt{2} \\
 &\text{而 } x e^{-x^2} \text{ 为奇函数}
 \end{aligned}$$

$$\begin{aligned}
 \therefore \int_{-\infty}^{+\infty} x e^{-\frac{(x-a)^2}{2b^2}} dx &= \frac{\sqrt{2}}{2} \cdot 6a \\
 \therefore \mu &= a
 \end{aligned}$$

3. 判断下列积分的敛散性:

- | | |
|--|---|
| (1) $\int_1^{+\infty} \frac{dx}{x^2 + \sqrt[3]{x^4 + 3}}$; | (2) $\int_2^{+\infty} \frac{dx}{2x + \sqrt[3]{x^2 + 1} + 6}$; |
| (3) $\int_0^2 \frac{dx}{\sqrt[3]{x} + 3\sqrt[4]{x} + x^3}$; | (4) $\int_0^1 \frac{dx}{\sqrt[3]{1-x^4}}$; |
| (5) $\int_0^1 \frac{\sin x}{x^{3/2}} dx$; | (6) $\int_0^{+\infty} \frac{\sin x}{x} dx$; |
| (7) $\int_1^{+\infty} x^a e^{-x^2} dx \quad (a > 0)$; | (8) $\int_0^1 \frac{\ln x}{1-x} dx$; |
| (9) $\int_0^{\pi/2} \frac{dx}{\sin^2 x \cos^2 x}$; | (10) $\int_0^{+\infty} \frac{dx}{x^p + x^q} \quad (p \geq 0, q \geq 0)$. |

(1) 收敛

(3) 收敛.

$$(5) \quad \lim_{x \rightarrow 0} \frac{\sin x}{x^{3/2}} = 1, \quad \text{收敛.}$$

$$(6) \quad \lim_{x \rightarrow 0} \frac{\sin x}{x} = 1, \quad \therefore 0 \text{ 不是瑕点}$$

$\therefore \frac{1}{x} \downarrow$ 且 $\rightarrow 0, \quad \forall A > 0, \quad \left| \int_0^A \sin x dx \right| \leq 2$, 有界
由狄利克雷判别法级数收敛.

$$(7) \quad \lim_{x \rightarrow \infty} \frac{x^a e^{-x^2}}{e^{-\frac{x^2}{2}}} = 0, \quad \text{而 } \int_1^{+\infty} e^{-\frac{x^2}{2}} dx \text{ 收敛, } \therefore \text{收敛}$$

$$(9) \quad \int_0^{\frac{\pi}{2}} \frac{d \tan x}{\sin^2 x} = \int_0^{\frac{\pi}{2}} \frac{1 + \tan^2 x}{\tan^2 x} d \tan x = \int_0^{+\infty} \frac{1+t^2}{t^2} dt = \int_0^{+\infty} \left(1 + \frac{1}{t^2} \right) dt = \left(t - \frac{1}{t} \right) \Big|_0^{+\infty}, \quad \lim_{t \rightarrow \infty} \left(t - \frac{1}{t} \right) \text{ 不存在, } \therefore \text{发散}$$

4. 判别下列积分是绝对收敛还是条件收敛：

$$(1) \int_0^{+\infty} \frac{\sqrt{x} \cos x}{x+3} dx; \quad (2) \int_1^{+\infty} \frac{\cos(3x+2)}{\sqrt{x^3+1} \sqrt[3]{x^2+1}} dx.$$

5. 叙述关于瑕积分的狄利克雷判别法及阿贝尔判别法.

6. 设 $f(x)$ 定义在 $(a, b]$ 上并以 a 为瑕点. 试将瑕积分

$$\int_a^b f(x) dx$$

$$4. (1) \int_0^{+\infty} \left| \frac{\sqrt{x} \cos x}{x+3} \right| dx > \int_{n^2}^{(n+1)^2} \left| \frac{\sqrt{x} \cos x}{x+3} \right| dx$$

1. 求下列函数的极限：

$$(1) \lim_{k \rightarrow 0} \int_0^{\pi/2} \frac{d\varphi}{\sqrt{1-k^2 \sin^2 \varphi}}; \quad (2) \lim_{k \rightarrow 1-0} \int_0^{\pi/2} \sqrt{1-k^2 \sin^2 \varphi} d\varphi;$$

$$(3) \lim_{\alpha \rightarrow 0} \int_{\alpha}^{\alpha+\pi/2} \frac{\sin^2 x}{4+\alpha x^2} dx; \quad (4) \lim_{\alpha \rightarrow 0} \int_0^{1+\alpha} \frac{dx}{1+x^2+\alpha^2};$$

$$(5) \lim_{y \rightarrow 0} \int_0^1 \frac{e^x \sin xy}{y+1} dx.$$

$$1. (1) f(\varphi) = \frac{1}{\sqrt{1-k^2 \sin^2 \varphi}} \text{ 在 } \mathcal{I} \text{ 上连续}$$

$$\lim_{k \rightarrow 0} \int_0^{\frac{\pi}{2}} \frac{d\varphi}{\sqrt{1-k^2 \sin^2 \varphi}} = \int_0^{\frac{\pi}{2}} \lim_{k \rightarrow 0} \frac{1}{\sqrt{1-k^2 \sin^2 \varphi}} d\varphi = \frac{\pi}{2}$$

$$(2) 1$$

$$(3) \lim_{\alpha \rightarrow 0} \int_{\alpha}^{\alpha+\frac{\pi}{2}} \frac{\sin^2 x}{4+\alpha x^2} dx = \int_0^{\frac{\pi}{2}} \frac{\sin^2 x}{4} dx = \frac{\pi}{16}$$

$$(4) \lim_{\alpha \rightarrow 0} \int_0^{1+\alpha} \frac{dx}{1+x^2+\alpha^2} = \int_0^1 \frac{dx}{1+x^2} = \arctan 1 = \frac{\pi}{4}$$

$$(5) 0$$

2. 求下列函数的导函数:

$$(1) g(y) = \int_{a-ky}^{a+ky} f(x) dx, \quad \text{其中 } f(x) \text{ 在 } (-\infty, +\infty) \text{ 上连续};$$

$$(2) g(y) = \int_{\sin y}^{\cos y} e^{y\sqrt{1-x^2}} dx, \quad 0 \leq y \leq \frac{\pi}{2};$$

$$(3) g(y) = \int_0^y \frac{\ln(1+xy)}{x} dx, \quad 0 < y < +\infty;$$

$$(4) g(y) = \int_0^{y^2} \sin(x^2 + y^2) dx, \quad -\infty < y < +\infty.$$

$$(1) \quad g'(y) = 0 + k \cdot f(a+ky) + k f(a-ky) \\ = k [f(a+ky) + f(a-ky)]$$

$$(2) \quad \frac{\partial f}{\partial y} = \sqrt{1-x^2} \cdot e^{y\sqrt{1-x^2}}, \text{ 在定义域内连续}$$

$$g'(y) = \int_{\sin y}^{\cos y} \sqrt{1-x^2} \cdot e^{y\sqrt{1-x^2}} dx - \sin y \cdot e^{\cos y \sqrt{1-x^2}} - \cos y \cdot e^{\sin y \sqrt{1-x^2}}.$$

$$(3) \quad \frac{\partial f}{\partial y} = \frac{1}{1+xy}, \text{ 在 } (0, +\infty) \text{ 上连续}$$

$$g'(y) = \int_0^y \frac{1}{1+xy} dx + \frac{\ln(1+y^2)}{y} = \frac{1}{y} \ln(1+xy) \Big|_0^y + \frac{\ln(1+y^2)}{y} = \left(\frac{2}{y}\right) \ln(1+y^2)$$

$$(4) \quad \frac{\partial f}{\partial y} = 2y \cdot \cos(x^2 + y^2), \text{ 在 } \mathbb{R} \text{ 上连续.}$$

$$g'(y) = \int_0^{y^2} 2y \cos(x^2 + y^2) dx + 2y \cdot \sin(y^2 + y^2)$$

3. 利用积分号下求导数的方法求下列积分:

$$(1) g(a) = \int_0^{\pi/2} \frac{\arctan(a \tan x)}{\tan x} dx, \quad -\infty < a < +\infty \quad (\text{提示: 先求 } g'(a));$$

$$(2) g(a) = \int_0^{\pi/2} \ln \frac{1+a \cos x}{1-a \cos x} \cdot \frac{dx}{\cos x}, \quad -1 < a < 1;$$

$$(3) I(a) = \int_0^{\pi/2} \ln(a^2 \sin^2 x + \cos^2 x) dx, \quad a \neq 0.$$

$$g'(a) = \int_0^{\pi/2} \frac{\tan x}{\tan x} \cdot \frac{1}{1+a^2 \tan^2 x} dx = \int_0^{\pi/2} \frac{1}{1+a^2 \tan^2 x} dx$$

$$\text{设 } \tan x = t, \quad t \in [0, +\infty)$$

$$g'(a) = \int_0^{+\infty} \frac{1}{1+a^2 t^2} \cdot \frac{1}{1+t^2} dt$$

$$= \int_0^{+\infty} \left(\frac{\frac{1}{1+a^2 t^2}}{1+t^2} - \frac{1}{1+t^2} \right) dt$$

$$= \frac{1}{a^2-1} \left(a \arctan \frac{t}{a} - \arctan t \right) \Big|_0^{+\infty}$$

$$= \frac{1}{a^2-1} (a-1) \frac{\pi}{2} = -\frac{\pi}{2} \cdot \frac{1}{a+1}$$

$$\therefore g(a) = \frac{\pi}{2} \ln |a+1|$$

$$\frac{1}{1+a^2 t^2} - \frac{1}{1+t^2} = \frac{(1-a^2)t^2}{(1+a^2 t^2)(1+t^2)}$$

$$\frac{a^2}{1+a^2 t^2} - \frac{1}{1+t^2} = \frac{a^2 + a^2 t^2 - 1 - a^2 t^2}{(1+a^2 t^2)(1+t^2)} = \frac{a^2-1}{(1+a^2 t^2)(1+t^2)}$$

$$(2) \quad g'(a) = \int_0^{\pi/2} \frac{1}{\cos x} \left(\cos x \cdot \frac{1}{1+a \cos x} + \cos x \cdot \frac{1}{1-a \cos x} \right) dx$$

$$= \int_0^{\pi/2} \left(\frac{1}{1+a \cos x} + \frac{1}{1-a \cos x} \right) dx$$

$$= \int_0^{\pi/2} \frac{2}{1-a^2 \cos^2 x} dx$$

$$= 2 \cdot \int_0^{\pi/2} \frac{\sec^2 x}{\sec^2 x - a^2} dx = 2 \cdot \int_0^{+\infty} \frac{dt}{t^2 + 1 - a^2} dt = 2 \cdot \frac{1}{\sqrt{1-a^2}} \arctan \frac{t}{\sqrt{1-a^2}} \Big|_0^{+\infty}$$

$$= \frac{\pi}{\sqrt{1-a^2}}$$

$$\therefore g(a) = \pi \arcsin a$$

$$(3) I(a) = \int_0^{\pi/2} \ln(a^2 \sin^2 x + \cos^2 x) dx, \quad a \neq 0.$$

4*. 求积分 $\int_0^1 \frac{\arctan x}{x \sqrt{1-x^2}} dx$ (提示: $\frac{\arctan x}{x} = \int_0^1 \frac{dy}{1+x^2 y^2}$).

5*. 设 $f(x)$ 在 $[0, 1]$ 上连续, 证明: 对任意 $\alpha > 0$, $\alpha < 1$, 成立下式

$$(3) I'(a) = \int_0^{\frac{\pi}{2}} 2a \sin^2 x \cdot \frac{1}{a^2 \sin^2 x + \cos^2 x} dx$$

$$= \int_0^{\frac{\pi}{2}} \frac{2a \tan^2 x}{a^2 \tan^2 x + 1} dx, \quad \tan x = t, \quad x = \arctan t.$$

$$= 2a \int_0^{+\infty} \frac{t^2}{a^2 t^2 + 1} \cdot \frac{1}{1+t^2} dt$$

$$= 2a \cdot \int_0^{+\infty} \left(\frac{1}{a^2 t^2 + 1} - \frac{1}{a^2 t^2 + 1} \cdot \frac{1}{1+t^2} \right) dt$$

$$= 2a \cdot \left(\frac{1}{a} \arctan(at) - \frac{1}{a^2-1} \left[a^2 \frac{1}{a} \arctan(at) - \arctan t \right] \right) \Big|_0^{+\infty}$$

注意正负号的影响
 $a > 0$ 与 $a < 0$, 分类讨论

$$= 2a \cdot \left(\frac{\pi}{2} \cdot \frac{1}{a} - \frac{1}{a^2-1} \cdot \frac{\pi}{2} \cdot (a-1) \right)$$

$$= \pi - \frac{\pi}{2} \cdot 2a \cdot \frac{1}{a+1}$$

$$= \frac{1}{a+1} \pi, \quad \underline{I(1) = 0.} \text{ (初值条件)}$$

$$I(a) = \pi \ln |a+1| + C, \quad I(1) = \pi \ln 2 + C = 0, \quad C = -\pi \ln 2$$

$$I(a) = \pi \cdot \ln \frac{|a+1|}{2}$$

4*. 求积分 $\int_0^1 \frac{\arctan x}{x\sqrt{1-x^2}} dx$ (提示: $\frac{\arctan x}{x} = \int_0^1 \frac{dy}{1+x^2y^2}$).

5*. 设 $f(x)$ 在 $[a, b]$ 上连续. 证明: 对任意一点 $c, a < c < b$, 成立下列等式:

$$\begin{aligned}
 4. \quad \int_0^1 \frac{\arctan x}{x\sqrt{1-x^2}} dx &= \int_0^1 \left(\int_0^1 \frac{1}{(1+x^2y^2)\sqrt{1-x^2}} dy \right) dx \\
 \int_0^1 \frac{dx}{(1+x^2y^2)\sqrt{1-x^2}} &= \int_0^1 \frac{d\arcsin x}{1+x^2y^2} = \int_0^{\frac{\pi}{2}} \frac{dt}{1+y^2\sin^2 t} = \int_0^{\frac{\pi}{2}} \frac{\sin^2 t + \cos^2 t}{\sin^2 t + \cos^2 t + y^2\sin^2 t} dt = \int_0^{\frac{\pi}{2}} \frac{\tan^2 t + 1}{(y^2+1)\tan^2 t + 1} dt = \int_0^{\frac{\pi}{2}} \frac{1}{(y^2+1)\tan^2 t + 1} d(\tan t) \\
 \arcsin x &= t, \quad x = \sin t \\
 &= \int_0^{+\infty} \frac{1}{(y^2+1)t^2+1} dt \\
 &= \left[\frac{1}{y^2+1} \cdot \frac{1}{\sqrt{y^2+1}} \cdot \arctan(\sqrt{y^2+1} \cdot t) \right] \Big|_0^{+\infty} \\
 &= \frac{\pi}{2} \cdot \frac{1}{\sqrt{y^2+1}} \\
 \therefore \text{原式} &= \frac{\pi}{2} \int_0^1 \frac{1}{\sqrt{y^2+1}} dy \\
 &= \frac{\pi}{2} \cdot \left(\ln|y + \sqrt{y^2+1}| \right) \Big|_0^1 = \frac{\pi}{2} [\ln(\sqrt{2}+1)] \\
 &= \frac{\pi}{2} \ln(\sqrt{2}+1)
 \end{aligned}$$

5*. 设 $f(x)$ 在 $[a, b]$ 上连续. 证明: 对任意一点 $c, a < c < b$, 成立下列等式:

$$\lim_{h \rightarrow 0} \frac{1}{h} \int_a^c [f(t+h) - f(t)] dt = f(c) - f(a)$$

$$\int_a^c f(t+h) dt = \int_{a+h}^{c+h} f(t) dt$$

$$\lim_{h \rightarrow 0} \int_a^c [f(t+h) - f(t)] dt = \lim_{h \rightarrow 0} \left[\int_{a+h}^{c+h} f(t) dt - \int_a^c f(t) dt \right] = 0$$

$$\lim_{h \rightarrow 0} \frac{\int_{a+h}^{c+h} f(t) dt - \int_a^c f(t) dt}{h} = \lim_{h \rightarrow 0} [f(c+h) - f(a+h)] = f(c) - f(a)$$

